

Dyania Health Hackathon, Athens, September 2026

Dynamic prediction of bioprosthetic aortic valve failure to guide personalised surveillance

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17 September 2026

Structural Valve Deterioration

Permanent leaflet change: calcification, tearing, fibrous ingrowth. The only remedy is another procedure, and neither route is free of harm.

TEN-YEAR OUTCOMES, NOTION TRIAL

20.8%

surgical valves deteriorated

15.4%

transcatheter, same trial

62.7%

died before any of it

HOW LATE THEY ARE FOUND: 3 TO 9 MONTHS PAST THE WINDOW

At the moment of	The ideal window	Where we actually find them
NYHA class	I or II, asymptomatic	III or IV, dyspnoea at rest
Presentation	stable haemodynamics	decompensated heart failure or shock
Left ventricle	normal size, preserved LVEF	irreversible dilation, pulmonary HTN
Procedure	elective, planned in advance	urgent or emergent, within 24 to 48 hours

NOTION: Thyregod HGH et al., Eur Heart J 2024;45:1116-1124. The ideal window and the state at detection are a clinical synthesis of published reviews and case series, not a cohort measurement

Surveillance today

Deterioration is only ever *seen* at an echo, so the recorded date is the date somebody looked, not the date the valve began to fail.

HOW IT WORKS TODAY

SAVR: first echo at 1 year. TAVR: often at 6 months. Then annually for both, whatever the patient's risk.

WHERE IT BREAKS

40%

of timely surveillance echos are never performed

WHAT LATE COSTS: OPERATIVE MORTALITY

Elective redo AVR

2.5%

Urgent or emergent redo AVR

4.6 to 17%

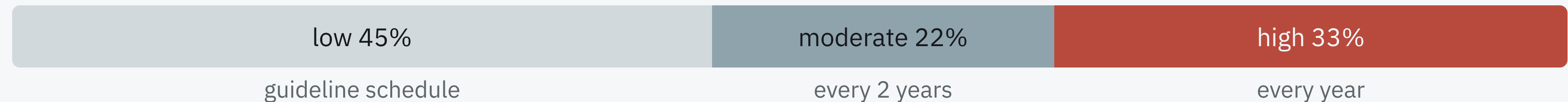
Finding it late carries a **50% higher** long-term risk of death, and costs about **\$36,740** more per patient over three years.

40%: Circulation 2026, a native aortic stenosis cohort. Redo AVR mortality: published surgical series. Cost and death-risk figures: United States estimates, citations pending

Hypothesis

Durability risk is not uniform, and it is knowable from what the hospital already records.
Estimate it at every visit.

Of every 100 visits



Of every 100 failures



A third of the visits carry three fifths of the failures. That mismatch is the opportunity.

1,000 → 483 echos per 1,000 patient-years: an annual study for everyone, against the risk-guided schedule

A visit is one decision about when to image next, so the bands are visits, not patients. Tier shares and workload from *model/decision_curve.md*, synthetic cohort. Observed 5-year risk inside the tiers: 7.6%, 12.9% and 28.1%

Study design

Population

- Adults with a bioprosthetic aortic valve (SAVR or TAVR), alive and failure-free at the reference echo, 30 to 90 days after implant.
- Risk re-estimated at every visit: 6 months, 1 year, then yearly to 10 years.
- *Excluded*: mechanical valves, a second prosthetic valve, non-structural dysfunction at baseline.

Endpoints, VARC-3

- Stage 2 or worse deterioration against the patient's own reference echo, or reintervention for structural failure, within 5 years (also 2 and 8).
- Death is a competing event, not censoring.

What is not deterioration

- Endocarditis, thrombosis, paravalvular leak: censored, not counted. 8 of 18 candidate events in the extract.
- Patient–prosthesis mismatch is a risk factor, not an event.

Data sources

Source	What we take from it	Realistic in deployment
Operative reports	valve model, size, SAVR or TAVR	Yes, as free text
Echo reports	gradients, DVI, regurgitation, LVEF, date	In the echo system, or as report text
Clinical notes	diabetes, kidney disease, smoking	Yes, but written mostly when present
Demographics	age, sex	Yes (age was redacted in our extract)
Labs and medications	renal function, anticoagulation	Yes, fully structured
Vital status	the competing risk	Needs a death-registry link

HOW WE READ THE NOTES

Rules plus LLM agents: fixed fields, each with the sentence it came from.

SUPPLIED EXTRACT, MEASURED AGAINST THAT SCHEMA

117

patients, dates in years

1.62

echos each

16%

more than one gradient

0

deaths recorded

10

events in 51 episodes

The extract cannot train or test a model on its own. Models are trained, validated and tested on a synthetic cohort built from published evidence, and the extract is only scored.

Validation

How we validate

- **Synthetic train, validation, test.** 6,000 patients.
- **Split.** Temporal at 2018, grouped by patient: no patient crosses it. 2018 is where case mix turns, 88% of later implants are TAVR.
- **Validation.** Tuning and recalibration by cross-validation inside the training set only.
- **Metrics.** Competing-risk time-dependent AUC at 2, 5 and 8 years, calibration against Aalen–Johansen observed risk.
- **Comparators.** Valve age alone, the guideline calendar, a Cox model on published risk factors, and a regression baseline.

RESULTS, 5-YEAR AUC, MEAN OF 5 DRAWS

Model	All visits	First visit
Gradient boosting	0.76	0.60
Regression baseline	0.76	0.61
Valve age only	0.73	0.50

0.60

at the first visit, where valve age alone gives 0.50

6% vs 6.6%

predicted vs observed risk at the first visit

From the first visit the question can be answered modestly; prediction improves as serial echos accumulate. The real extract lacks ages, dated echos and deaths, and that is the gap the protocol closes.

Model

A competing-risks survival model, re-run at every visit: the chance of failure in each follow-up year, chained into risk at 2, 5 and 8 years.

Why this model

- **Survival, not a fixed-horizon classifier.** It uses every observed year, even for patients followed for less than five.
- **Not a sequence model.** Most patients have one or two echos, so a landmark model uses the history known at each visit without needing long sequences.
- Primary: gradient boosting. Baseline: logistic regression.

How we chose the features

- **1. Clinical input.** Our clinician named the factors; we built all 42 the data can supply.
- **2. Screening, built and rejected.** Automatic selection was unstable across draws.
- **3. Tested, then fixed.** The clinician-aligned 21 won; scan frequency was dropped.

From risk to action

5-year risk	Next echo
under 5%	guideline
5% to 15%	every 2 years
15% or more	every year

The clinician's factors earn their place: on full-quality synthetic data they lift the first-visit 5-year AUC from 0.54 to 0.62 (boosted) and 0.55 to 0.64 (baseline), one cohort. Acting on the model beats scanning everyone and changing nothing from 5% risk upward

Results

COMPETING-RISK AUC, FULL-QUALITY SYNTHETIC, 5 DRAWS

Model	2 years	5 years	8 years
Gradient boosting	0.83	0.76	0.76
Regression baseline	0.81	0.76	0.76
Valve age only	0.79	0.73	0.71
Guideline calendar	0.69	0.67	0.58

Every risk model beats valve age at every horizon, by 0.02 to 0.06, with a spread across draws of about 0.01.

Predicted vs observed 5-year risk: 18% vs 15%.

REAL EXTRACT, SCORED ONLY, NEVER TRAINED ON

0.72 regression, 0.69 boosting, 0.72 valve age, against 16.6% observed. With 10 events no model is shown to rank better than valve age alone, and we say so.

What drives the risk, SHAP

- 1. years since implant
- 2. SAVR or TAVR
- 3. age, younger means higher risk
- 4. body surface area
- 5. change in mean gradient
- 6. Trifecta valve

Next steps

What it enables

1. Fewer scans than yearly screening, focused on the patients who fail.
2. Earlier attention for high-risk patients.

What validation needs

1. More data: 3,580 patients followed 5 years (about 565 failures).
2. Complete data: dated echos, age, death records.
3. Confirmed labels: two blinded cardiologists per failure.

Given three months

1. One hospital's full cohort, linked to death records.
2. Extraction checked against a cardiologist's reading.
3. The clinician's remaining factors: blood pressure, cholesterol, labs, medications.
4. **Fine-tune and compare more models.**

Deployed in a closed hospital environment: no patient data leaves the site, and the model runs on standard hospital hardware.